# 1. Loss and score

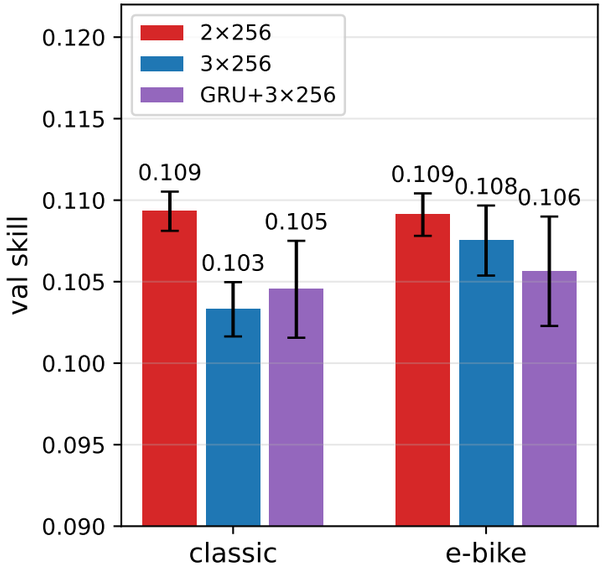
- Loss: mean squared error (MSE) of the future count of each bike type.
- Score: skill = 1 − MSE_model / MSE_persistence, because a raw MSE has no reference.
- Persistence = doing nothing: predict that the count now stays the same.
- Skill 0 = no better than doing nothing; 1 = perfect.
- We pick models by their skill on the val days (Oct 19–23).

# 2. What works, and why

**Pick: MLP 2×256 + dropout 0.1:  +0.006 / +0.007 val skill over the Nov 10 base (≈ 2.3–2.6 SE)**

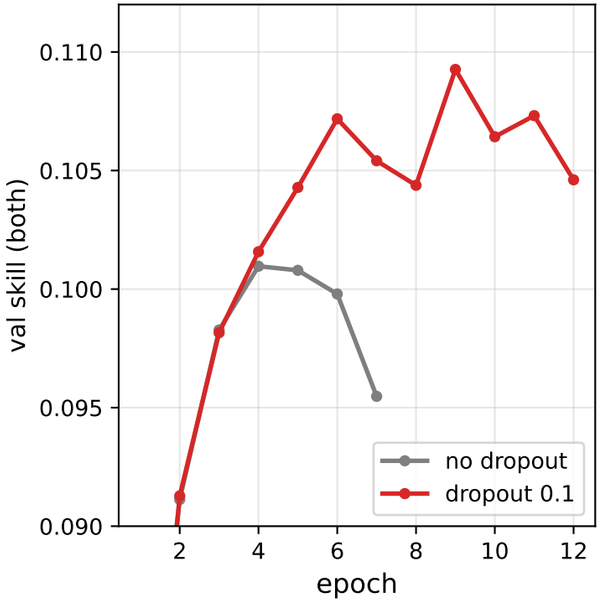
Mean ± sd over seeds 0–2 on val; each panel title is the claim, the line below it the reason

### Architecture: a 2-layer MLP. Deeper or a GRU add nothing.



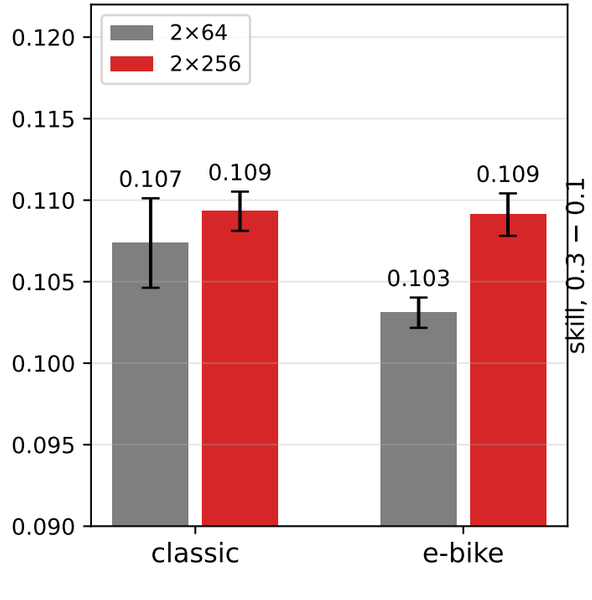

No LSTM: the table has only 3 lag steps (t−30, t−10, t).

### Dropout is the main gain.



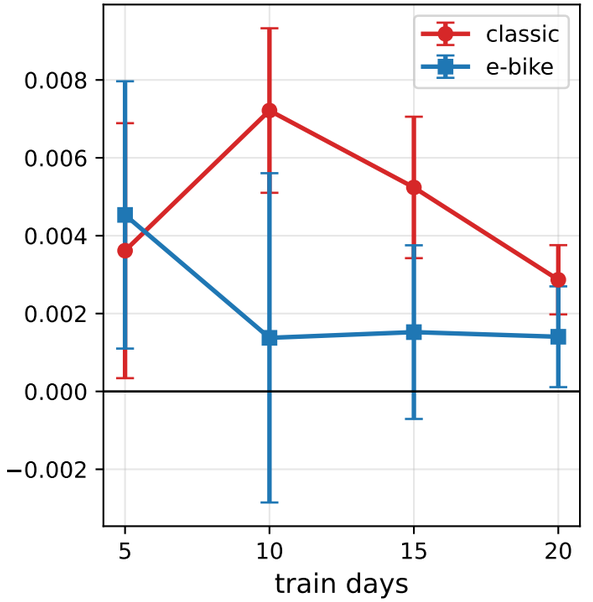

Without it, 2×256 overfits after epoch 4–6; with it, it keeps learning.

### Width (64 → 256) helps e-bikes only.




On classic bikes, 2×64 + dropout is within noise of 2×256.

### Why not dropout 0.3, which looks better?

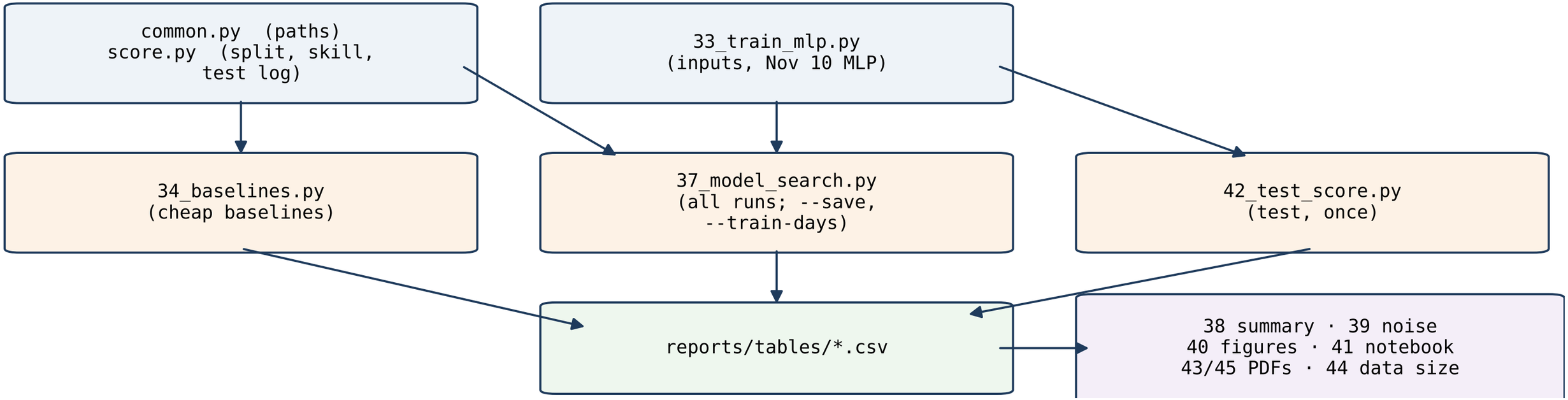



Lead +0.003 / +0.001: noise except classic all rows; nil on low rows; 2× the epochs; not growing with data.

Sources: model_search_summary.csv, model_search_curves.csv, val_noise_gap.csv, data_size_gap.csv (src/38, 39, 44)

# 3. Code: verify it, change it

All in src/ of the project repo · <GitHub link: to be added (repo not decided yet)>



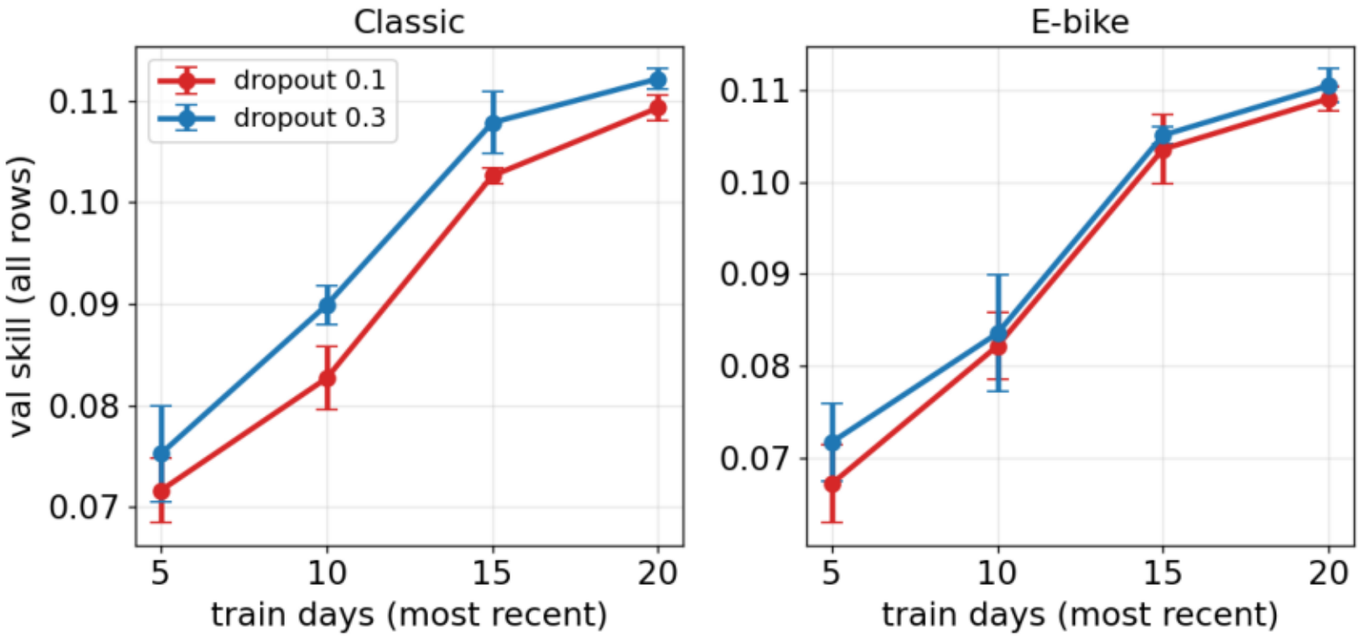

- Reproduce (from src/, Python env from requirements.txt): python 37_model_search.py --step 1 (or --run NAME --budget full --seed 0 1 2); python 38_search_report.py --finalists …; python 39_val_noise.py; python 44_data_size.py; python 42_test_score.py --dry (val checks only).
- Trace: notebooks/model_search_trace.ipynb (+ .html) prints every number of MODEL_SEARCH.md from its CSV with the script and commit; it checks the data SHA-1 and that the search never reads the test split.
- To change the model: add a config to GRIDS in 37_model_search.py; every run is logged to model_search_runs.csv. Data path: score.TEAM.

# Backup

B1  More data  ·  B2  Test result

# B1. More data helps more than any architecture change

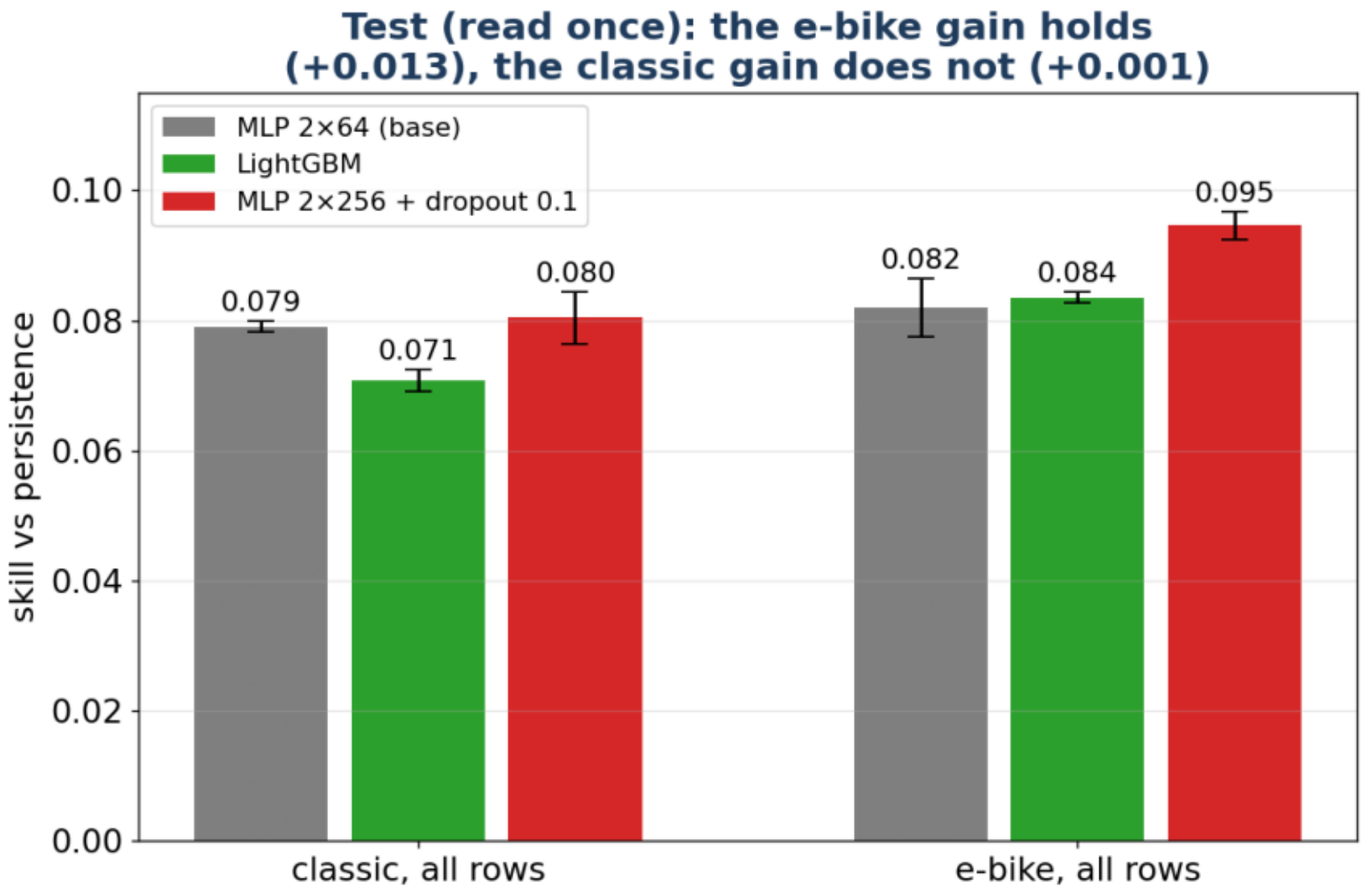
**More train days still help: 15 → 20 days adds +0.007 / +0.006 (≈ +0.009 / +0.008 per week), but the gain is slowing**




Mean ± sd over seeds 0–2; same val days (data_size_summary.csv, 44_data_size.py)

- Refit on the most recent 5 / 10 / 15 / 20 train days, same val days, same updates per epoch.
- 15 → 20 days still adds +0.007 / +0.006 (≈ +0.009 / +0.008 per week): the model is data-limited, though the gain is slowing.
- Compare: the whole architecture and dropout search gained +0.006 / +0.007.
- The pollers kept running after Oct 28, so a later table build adds training days.

# B2. Test result (read once)




Test Oct 24–28, mean ± sd over seeds 0–2 (test_team_summary.csv, 42_test_score.py)

- Chosen vs Nov 10 base on test: +0.001 classic, +0.013 e-bike.
- Test starts on a weekend, val has none: chosen classic 0.034–0.061 on Sat/Sun vs 0.114–0.117 on Mon/Tue.
- Protocol fixed and committed before reading; 3 seeds; nothing tuned after.